Instructions for Using the BiasFix App

Version 1.0 beta

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1 Overview

The BiasFix app is designed to standardise scores to adjust for examiner bias, especially in oral examinations where subjective judgments can introduce variability. This app enables users to normalise scores across different examiners, ensuring more equitable student assessments. The app provides templates for examination data formatting, facilitates data upload, visualises both raw and adjusted scores, and analyses data using linear mixed effects models (LMMs) to account for examiner-specific variability.

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2 Instructions

2.1 Downloading Templates and Sample Data

2.1.1 Download Formatted Templates

The app provides pre-formatted templates that allow you to easily structure your data. Select the format you prefer:

- **CSV Template:** Click on "Formatted .csv Template" to download a CSV file.
- **XLS Template:** Click on "Formatted .xls Template" to download an Excel file (97-2004 format).
- **XLSX Template:** Click on "Formatted .xlsx Template" to download a modern Excel file.

Each template contains the following essential columns: - `Student_ID`: A unique identifier for each student. - `Examiner`: The name or code of the examiner who evaluated the student. - `Score`: The score assigned by the examiner.

2.1.2 Download Sample Dataset

To explore how the app works before uploading your own data, download the sample dataset by clicking on "Sample Dataset". This sample file follows the same format as the templates and can be used for testing the app's features.

2.2 Preparing and Uploading Data

2.2.1 Format Your Data

Ensure your dataset is structured with the following columns:

- `Student_ID`: Unique identifier for each student.
- `Examiner`: The name or code of the examiner.
- `Score`: The raw score assigned to the student.

The app accepts CSV, XLS, and XLSX formats. Any data missing in the `Score` column should be removed before uploading to prevent processing errors.

2.2.2 Upload Your Data

- Go to the "Standardisation" tab.
- Click the "Upload CSV/XLS/XLSX File" button to upload your data file.
- The app will automatically read the data and prepare it for analysis.

2.2.3 Input Total Score

To ensure accurate scaling of scores, input the maximum possible score (e.g., 100) in the "Input Total Score" field.

2.3 Viewing and Downloading Results

2.3.1 Visualisations

The app provides several visual tools to help you understand your data before and after standardisation:

- **Original Boxplot:** Displays the distribution of raw scores per examiner. Download this plot by clicking the "Download Original Boxplot" button.
- **Retransformed Boxplot:** Shows the distribution of scores after adjusting for examiner bias. The dashed blue line indicates the mean score across all examiners. Download it using the "Download Retransformed Boxplot" button.
- **Histograms:** Visualise the frequency distribution of both raw and retransformed scores. These histograms provide an overview of the spread and central tendencies of your scores. View them interactively within the app.

2.3.1.1 Interaction with Plots The visualisations allow for various interactive features: - **Zoom and Pan:** Click and drag to zoom in on specific sections of the plot. Use panning to shift your view across different areas. - **Hover for Summary Statistics:** Hovering over the boxplots or histograms reveals summary statistics for each examiner, including median, quartiles, and outlier counts. - **Outliers:** Outliers are calculated as points beyond 1.5 times the interquartile range (IQR) from the lower and upper quartiles.

2.3.2 Summary Statistics

Summary statistics of the retransformed scores, such as mean, standard deviation, and other descriptive measures, are available for download. Click on "Download Summary Statistics" to obtain this data.

2.3.3 Modified Data

After standardisation, the modified data, including the adjusted scores, can be downloaded by clicking on "Download Modified Data". The column `Retransformed_Score` represents the final adjusted score for each student.

2.4 Recommendations for Data Collection

To ensure the reliability of the linear mixed effects model, it is recommended that each examiner assesses at least 30 students. This provides sufficient data to accurately estimate examiner-specific effects and improves the robustness of the standardisation process.

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3 Statistical Techniques

3.1 Z-Score Standardisation

The BiasFix app uses **Z-score standardisation** to adjust for examiner bias. Z-scores represent the number of standard deviations a score is from the mean. The formula for calculating the Z-score is:

$$Z_i = \frac{X_i - \mu_j}{\sigma_j}$$

Where: - Z_i is the Z-score for the score X_i. - X_i is the raw score. - μ_j is the mean score for examiner j. - σ_j is the standard deviation of scores for examiner j.

This transformation normalises scores across different examiners, adjusting for variability in their scoring tendencies.

3.2 Retransformation of Scores

After standardisation, the Z-scores are retransformed to the original scale using the overall mean and standard deviation of all scores combined. The retransformation formula is:

$$Y_i = Z_i \times \sigma_{\text{overall}} + \mu_{\text{overall}}$$

Where: - Y_i is the retransformed score. - Z_i is the standardised Z-score. - σ_{overall} is the standard deviation of all raw scores across examiners. - μ_{overall} is the mean of all raw scores across examiners.

By retransformation, the scores are brought back to the original scoring scale, while accounting for examiner bias.

3.3 Linear Mixed Effects Model

The app also uses a **Linear Mixed Effects Model (LMM)** to account for examiner-specific variability. The model has the following form:

$$Y_i = \beta_0 + u_j + \epsilon_i$$

Where: - Y_i is the score for student i. - β_0 is the fixed effect representing the overall mean score. - u_j is the random effect for examiner j, which accounts for examiner-specific deviations. - ϵ_i is the residual error term.

This model assumes that examiner effects are random, allowing each examiner to have their own baseline level of scoring. The LMM makes the final standardised scores fairer by controlling for systematic biases across examiners.

3.4 Random Intercepts

Random intercepts are the estimated deviations for each examiner from the overall mean score. These are derived from the LMM and are crucial for understanding how much an examiner's scores deviate from the average. You can download the random intercepts by clicking the "Download Random Intercepts" button.

3.5 Example of Statistical Adjustments

Consider an examiner who consistently scores students higher than others. The standardisation process adjusts these scores to be comparable with other examiners' scores, ensuring that a student's final score reflects their performance without examiner bias.

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4 Troubleshooting

4.0.1 Common Issues

- **Invalid File Format:** Ensure that your file is either in CSV, XLS, or XLSX format. Other formats will not be accepted.
- **Data Issues:** Double-check that your data is correctly formatted and that no critical information, such as student IDs or scores, is missing. Missing data may prevent the model from running correctly.

4.0.2 Additional Help

If you encounter issues not covered in this guide, please consult the "About" tab for more details or contact us directly at yasith.mathangasinghe1@monash.edu.
